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Chapter 19
The Road to Poker: Static Multi-Street Games

In the preceding chapters, we have considered a wide variety of games of the half-street and full-street varieties. One of the primary virtues that these games have is that they are relatively simple to solve; and games that we can solve can give us insight into the impact of changes to a particular dimension of the game. However, real poker is seldom played in half-street or full-street ways, and as we shall see in this chapter and the next, adding additional streets changes strategy in fundamental ways.

We begin by considering what we call ***static games.*** These are games where there are multiple streets of betting, but neither player's hand changes value. These games give us insight into how-bluffing works across multiple streets without considering the complexity of draws. After covering a few games of this type, we will move on to the more difficult and realistic games where draws are present.

The most basic static game of two streets that corresponds with our treatment of half street and full street games is the clairvoyant game.

Example 19.1 - The Two-Street Static Clairvoyant Game

Two full streets.
Static hand values.
Pot size of P bets.
Limit betting.
Y is clairvoyant.

Recall that in the one-street clairvoyant game, X never bet because such bets would never have positive value. Since Y was clairvoyant, he could exploit any such bets that X made with perfect information. The same situation occurs here. On either street, X will only check and call because Y can respond to any bet with perfect accuracy. This is the case here only because the game is static; that is, Y knows his ending hand value now. If there were draws where Y did not know the outcome, there might be situations where X would bet.

In the one street game, we found that the solution to the clairvoyrance game was that Y would bet all his winning hands, and bluff with a fraction a of his losing hands. X would respond by folding a of his hands, and calling with the rest. This was true except for the cases where Y had such a strong distribution of hands that he could bet all hands and X would be compelled to fold.

In the two-street game, we have an additional street of betting. This creates some additional strategic options for the two players. First consider Y's options.

Check both streets (strategy Υ_0)
Check the first street and bet the second (strategy Υ_1}
Bet the first street and check the second (equivalent to strategy Υ_1)
Bet both streets (strategy Υ_2)

Clearly Y will prefer to bet both streets with his nut hands; this maximizes value. Hands that employ strategies Υ_1 and Υ_0 will be bluffs, as well as some hands that employ strategy Υ_2. Y will also prefer to play the second version of Υ_1 where he bets the first street and checks the second. The check-first version is of course exploitable because X will know that Y does not have a nut hand because of his initial check.

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X's strategy must make Y indifferent between these three options when Y has a dead hand.
X has three options in response to Y's bets:

Fold immediately (strategy X_0)

Call the first street and fold the second (strategy X_1)

Call both streets (strategy X_2).

We have the following matrix (values from Y's perspective):

	X_0	X_1	X_2
Y_0	0	0	0
Y_1	+P	-1	-1
Y_2 (bluff)	+P	+ (P+ 1)	-2
Y_2 (value)	0	+1	+2

When Y uses strategy Y_2, we will call his fraction of value betting hands y_v and his fraction of bluffs y_b. We will call X's frequency of using his three strategies x_0, x_1, and x_2, and Y's frequency of using his three strategies y_0, y_1, and y_2.

Then we have the following indifference between Y_1 and Y_0:

$0 = Px_0 – x_1 - x_2$

We know that $x_1 + x_2 = 1 – x_0$, because the total frequencies for both players must sum to one. Hence:

$0 =1 – x_0 – Px_0$
$x_0 = 1/(1 +P) = \alpha$

So on the first street. X plays a strategy that is the same as his strategy in the one-street game. He folds α of his hands, and calls with the remainder.

Next, between Y_1 and Y_2 (bluff), we have:

$aP – x_1 - x_2 = aP+ (P+ 1)x_1 - 2x_2$
$(P+2)x_1 = x_2$

We have $x_1 + x_2 = 1 - a$, so:

$1 - a – x_1 = (P+ 2)x_1$
$x_1 = a_2(1 - a)$

On the second street, as well, X plays a strategy that is very much like his strategy in the single street game. After folding α of his hands on the first street, he has 1 - α hands left. The pot has grown by two bets because of the betting on the first street, so on the second street he now folds a_2 of his hands, and calls with the remainder. This is a very important idea:

The non-clairvoyant player plays each street independently as if it were a single street game.

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We now turn our attention to Y's strategy of betting and bluffing. Y must make X indifferent between his three strategies x_0, x_1, and x_2. We have the following indifference between x_0 and x_1.

$Py_1 + Py_by_2 = - y_1 + (P + 1)y_by_2 + y_vy_2$
$(P + 1)y_1 = y_vy_2 + y_by_2$
$y_1 = ay_2$

This indifference shows the second important result of this game.

The hands that Y bluffs once and then gives up on have the a bluffing ratio relationship to the total hands that Y will bet for value and bluff on the second street.

Next, the indifference between X_1 and X_2:

$-y_1 + (P + 1)y_by_2 + y_vy_2 = -y_1 – 2y_by_2 + 2y_vy_2$
$y_vy_2 = (P + 3)y_by_2$
$y_b = a_2y_v$

So on the second street, Y value bets and bluffs in the typical bluffing ratio out of epe hands he has left. Here the ratio is a_2 because the pot is two bets larger.

Let's recap this idea, because it is an important one. On the final street, Y will have a set of betting and bluffing hands appropriate to the pot size at that point. On the preceding street, he "pretends" that all those hands are value hands and bluffs appropriately to the current pot size. Then on the final street, he gives up on his "bluffs" from the first street, and only continues to bluff out of the "value hands" he carried forward from the last street.

This often leads to Y bluffing a larger number of hands than we saw in the one-street games. For example, suppose the pot is 2 bets before the first street and Y has just a small number of winning hands. Then we have the following:

On the river, the pot will be 4 bets. Y will want to value bet all his winning hands and $^1/_5$ as many bluffs on the river. So he "pretends" that his value bets plus another $^1/_5$ as many bluffs are value hands. On the first street, the pot is 2 bets, so he wants to bluff with his 'value hands" and $^1/_3$ as many dead hands. Since he has $^6/_5$ as many "value hands," he bluffs with the intention of giving up with $^2/_5$ as many hands as he has nut hands on the first street. The total number of dead hands he bets on the first street is $^3/_5$ as many as he has value hands.

He then bets all these hands, gives up on $^2/_3$ of his dead hands after the first street, and continues with $^1/_3$ of the dead hands and all the value bets on the second street. The second player cannot exploit this strategy, as we showed that he is indifferent between calling once, twice, or not at all.

This result generalizes to any number of streets and to any bet size, subject to the usual restrictions on strength of distribution. Suppose that we play the following game:

X and Y ante 2 units.
Both players receive hands, but both hands are revealed only to Y.
A complete board is dealt but revealed only to Y.
There are three streets - a flop where the bets are 1 unit, and a turn and river where they are 2 units.

If there's a showdown, the board and both hands are exposed and the showdown is as usual.

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Again in this game, X cannot profitably bet on any street; the solution is just as it was in the two-street game, except with an additional street and different bet sizes.

With winning hands, Y will bet on each street, and the pot size going into the river will be 4 + 2 + 4 = 10 units, with a bet size of 2. On the river, Y will bet with the $^1/_2$ of hands with which he will win, plus an additional $^2/_{12}$ as many, for a total of $^1/_{12}$. This makes a total of $^7/_{12}$ of his hands.

On the turn, the pot will be 6 bets, with a bet size of 2. So Y will have $^7/_{12}$of his hands designated 'Tor value," and will additionally bluff $^1/_4$ as many hands as that, or an additional $^7/_{48}$. So his total betting hands on the turn will be $^{35}/_{48}$.

On the flop, the pot will be 4 bets, with a bet size of 1. Y will take his $^{35}/_{48}$"value bets" and bluff an additional $^1/_5$ of that, or $^7/_{48}$. So his total betting hands on the flop will be $^{42}/_{48} = {}^7/_8$.

The solution is that Y bets $^7/_8$ of his hands total and gives up on $^1/_{12}$ of his dead hands after the flop and $^7/_{48}$ of his dead hands after the turn. X simply folds α of his hands on each street. The value of this game is easily found by assuming X folds immediately (since he is indifferent). 1/8 of the time, X wins the pot for a net to Y of 1.5 units. It's amazing that X can retain any equity at all in such a biased game.

We said that this extended to all bet sizes. In games where the clairvoyant player controls the bet size (such as no-limit), we can use the maximization techniques from previous chapters to find optimal bet sizes.

Example 19.2 - The Two-Street Static Clairvoyant No-Limit Game

Two full streets.
Static hand values.
Pot size of 1 unit.
No-limit betting (stacks of N units).
Y is clairvoyant.

We'll first consider the two-street case; the idea here generalizes to any number of streets. We saw in the previous game that the non-clairvoyant player plays each street as if it were a single-street game, folding a of his hands and calling with the remainder. Likewise, the clairvoyant player bets and bluffs using a-ratios to the hands he carries forward from street to street.

These relationships hold even when the bet sizes are not constant and can be determined by the clairvoyant player. As a result, we can find the clairvoyant player's equity in terms of bet sizes s_1 and s_2 and attempt to maximize this value. We have a matrix similar to the previous game, supposing the pot is 1 unit.

	X_0	X_1	X_2
Y_0	0	0	0
Y_1	+1	$-s_1$	$-s_1$
Y_2 (bluff)	+1	$+ (1+ s_1)$	$- (s_1 + s_2)$
Y_2 (value)	0	$+ s_1$	$+(s_1 + s_2)$

Writing the same indifference equations as in the previous game, we have:

$x_0 - s_1x_1 - s_1x_2 = 0$
$x_0 = s_1(1 - x_o)$

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$x_0 = s_1/(1+s_1)$

Again, we see that X folds α of his hands (recall that α = s/(l + s) as well as 1/(P+ 1)).

$x_0 + (1+s_1)x_1 - (s_1 + s_2)x_2 = x_0 - s_1x_1 - s_1x_2$
$(1+2s_1)x_1 = s_2x_2$
$x_2 = x_1(1+2s_1)/s_2$
$x_2+x_1 = x1(1+2s_1+s_2)/s_2$

Now let's consider what α_2 is for this game.

After the first street, the pot size will be $1 + 2s_1$ (the original pot, plus a bet and a call). So α_2 would be $s_2/(l + 2s_1 + s_2)$. Substituting this into the above equation, we have:

$x_2 + x_l = x_l / \alpha_2$
$x_1 = \alpha_2(x_2 + x_1)$

We saw this result previously. X plays each street as though it were a single street game, folding an appropriate a of his hands on each successive street. Of all the hands that call one bet (x_2 + x_1), X folds α_2 for the second bet.

The indifferences for X are likewise the same:

	X_0	X_1	X_2
Υ_0	0	0	0
Υ_1	+1	$-s_1$	$-s_1$
Υ_2 (bluff)	+1	$+ (1+ s_1)$	$- (s_1 + s_2)$
Υ_2 (value)	0	$+ s_1$	$+(s_1 + s_2)$

$y_1+y_2y^b = -s_1y_1+(1+s_1)y_2y^b+s_1y_2y_v$
$(1+s_1)y_1 = s_1y_2y^b + s_1y_2y_v$
$y_1 = y_2s_1 / (1+s_1)$
$y_1 = \alpha y_2$

Here we see, as before, that the number of hands that Y bluffs is equal to a times the number of hands he will both bluff and value bet on the river.

$-s_1y_1+(1+s_1)y_2y_b+s_1y_2y_v = -s_1y_1 - (s_1+s_2)y_2y_b +(s_1+s_2)y_2y_v$
$(1+s_1)y_2y_b+s_1y_2y_v = - (s_1+s_2)y_2y_b+(s_1+s_2)y_2y_v$
$(1+2s_1+s_2)y_2y_b = s_2y_2y_v$
$(1+2s_1+s_2)y_b = s_2y_v$
$y_b = \alpha_2y_v$

And on the river, the bluffing ratio is simply α_2. So this solution holds for any bet sizes (provided that the distributions are not so strong). To find the proper bet sizes s_1 and s_2, we maximize Y's equity for the two bet sizes.

We can make a nice simplification that additionally helps us understand the principle of optimal bet sizing. Suppose that Y bets s_1 into a pot of 1. If X were to call, then the pot would grow on this street from 1 to $(1 + 2 s_1)$. We define r_1 as this ratio; the ratio that the pot will grow if the bet s_1 is called, hi a similar fashion, we can define r_2 as the ratio by which the pot will grow if the bet s_2 is called, from $(1 + 2s_1)$ to $(1 + 2s_1 + 2s_2)$, or

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$(1 + 2s_1)/(1 + 2s_1 + 2s_2)$ and so on:
$r_n = (p_n+2s_n)/p_n$

Note that this *r* isn't necessarily the same as the previously discussed *r* (Equation 14.1).

We know from the above definition of r that for any street, the following is true:

$r = 1 + 2s$
(where 1 is the pot size coming in to this street).

From this, we can derive the following useful transformation:

$r – s = l +2s - s$
$r – s = l + s$
$r – s = (2 + 2s)/2$
$r – s = (l + r)/2$

We can also find the following:

$\alpha = s/(l +s)$
$\alpha = s/(r – s)$

Let M_n represent the amount of hands which are "for value," whether this means value bets on the next street or if this is the final street, hands which are the nuts. On any street, the amount of hands bet (bluffs and value bets) B_n for that street is:

$B_n = s_nM_n/(r_n - s_n) + M_n$
$B_n = M_n(r_n – s_n + s_n)/(r_n - s_n)$
$B_n = M_nr_n/(r_n – s_n)$
$B_n = 2M_nr_n/(l + r_n)$

This is an important result; it provides a "factor" that we can apply to the streets in order to find the number of bluffs on each street.

On the second street of our game, then, the total number of hands bet (if Y has y winning hands overall) will be:

$B_2 = 2yr_2/(l + r_2)$

while the hands bet on the first street will be:

$B_1 = 2B_2r_1/(1+r_1)$
$B_1 = 4yr_1r_2/(1+r_1)(1+r_2)$

Let's review all that math to distill it to a couple of results.

We defined values r_n that are the ratio of the pot after a bet and a call on that street to the pot before that street.

We found that for each street, we can express the total number of hands that are bet (bluffs or value bets) with the following formula:

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$B_n = 2M_n r_n/(l + r_n)$

where M is the number of hands that will be carried forward and bet on the next street (if there are streets remaining) or that are nut hands (if this is the final street).

If Y bets s_1 and s_2 at optimal frequencies, then X will be indifferent to calling or folding on any street. We can assume, therefore, that he simply folds immediately, which means that Y wins the pot with all his bluffs. We also know that X won't actually fold all his hands - in fact, he will call optimally. Since this is the case, Y will want to maximize the total amount he bets overall when he has a nut hand (and therefore also when he bluffs). As a result of this, we know that $s_1 + s_2$ will equal a constant of N; that is, Y will bet his entire stack on the two streets when he bets twice. Given that, we can calculate Y's equity by assuming that Y does not fold. Since the pot is one unit, Y maximizes his value by maximizing the fraction of hands that he bets.

The total number of hands Y bets is:

$$B_1 = \frac{4yr_1r_2}{(1+r_1)(1+r_2)}$$

The 4y term is of course a constant, so we need to maximize the following expression:

$r_1r_2/(1 + r_1)(1+r_2)$

r_1r_2 is a constant; we can see this as follows:

$r_1 = (l + 2s_1)$
$r_2 = (l+ 2s_1 + 2s_2)/ (1+2s_1)$
$r_2 = (1 + 2N)/ (1 + 2s_1)$
$r_1r_2 = 1 + 2N$

So we are left to maximize

$1/(1 + r_1)(1 + r_2)$

We know that both r_1 and r_2 are positive; hence, we can simply seek to minimize the denominator.

$(1+r_1)(1+r_2) = 1 + r_1 + r_2 + r_1r_2$
$(1+r_1)(1+r_2) = 2 + 2N + r_1 + r_2$

The constant terms fall out, and we are left to minimize $r_1 + r_2$ subject to the constraint that r_1 r_2 is constant. The arithmetic-geometric mean inequality shows us then that this expression is minimized when $r_1 = r_2$.

In fact this result generalizes to more than two streets. Using the same logic as we used above, we can find that the equity for Y for a three street game is:

$$B_1 = \frac{8yr_1r_2r_3}{(1+r_1)(1+r_2)(1+r_3)}$$

However if B_1 >=1, then what is happening is that Y does not have enough bluffs. Y should simply bet the growth of pot. And X should fold ***all*** hands.

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By reducing out terms as before, we are left to minimize the expression $r_1 + r_2 + r_3$ subject to the constraint that $r_1\ r_2\ r_3$ is a constant. This occurs when $r_1=r_2=r_3$. This result additionally applies to any number of streets.

We call this betting progression (growing the pot by the same amount on each street) ***geometric growth of pot***. This result is very significant to our understanding of no-limit bet sizing; in fact, as we will see in the next chapter, we use this as one of the bases of our entire bet-sizing scheme in no-limit holdem. When the game is static and one player is clairvoyant, the optimal bet size is to make the pot grow by the same amount on each street, such that on the last street the entire stack has been bet.

We will provide a short example game to contrast this strategy (against an optimal opponent) with another strategy that might seem reasonable but performs dramatically more poorly.

Example 19.3

Suppose we play the following game:

X and Y start with stacks of $185 and each antes $5.
Y draws a single card from a full deck.
There are three streets of no-limit betting.

If there is a showdown, then Y wins if his card is an ace or a king, and loses otherwise.

This game is equivalent to the clairvoyant game we discussed above, but with three streets. First, we can calculate the showdown value of this game; that is, suppose there were no betting. Then Y would win $^2/_{13}$ of the time for a total gross equity of $1.54 per play and a net loss of $3.46.

In order to make this game profitable, Y must make up $3.46 in equity from the betting. We will solve this game using the solutions we found earlier for two betsizing strategies:

Υ_1. Y bets one-third of his stack on each street.
Υ_2: Y bets the geometric growth of pot on each street.
For YI, of course, each bet will be $60. For Υ_2 we can take advantage of the fact that $10r_1r_2r_3 = 10r_3 = \$370$. So *r* is the cube root of 37, or about 3.3322.
On each street, $p_n + 2s_n = rp_n$

Street 1:
$\$10 + 2s_1 = \$10r$
$s_1 = \$11.66$

Street 2:
$\$33.32 + 2s_2 = \$33.32r$
$s_2 = \$38.86$

Street 3:
$\$111.04 + 2s_3 = \$111.04r$
$s_3 = \$129.48$

X will play his optimal strategy of calling $p_n/(p_n + s_n)$ of the time on each street. Doing this results in his strategy values as follows:

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Street	Y's bet (Υ_1)	X's call freq.	Y's bet (Υ_2)	X's call freq.
1	$60	14.29%	$11.66	46.2%
2	$60	68.42% -> 9.77%	$38.86	46.2% -> 21.3%
3	$60	80.6% -> 7.88°/o	$129.48	46.2% -> 9.84%
Total	**$60**	**7.88%**	**$90**	**4.06%**

We know that Y breaks even on his bluffs - on each street he bluffs appropriately to the number of hands he will carry over to the next street.

On his value hands, then, Y obtains the following value:

$< \Upsilon_1 >$ = (14.29%)($60) + (9.77%) ($60) + (7.88%)($60) =$19.16

$< \Upsilon_2 >$ = (46.17%) ($8.34) + (21.31%) ($22.26) + (9.84%) ($59.40) = $26.41

So Y does better by more than $7 per winning hand by betting the geometric growth. On a per-hand basis, this is enough to swing the game from negative (using Υ_1) to positive (using Υ_2). It is worth noting that the limit of ex-showdown value (no matter how much betting is left) must be equal to X's showdown value; if Y could extract more than that value from him, X would simply fold and surrender his $5 ante.

We also briefly look at a graph of the r constant against the stack sizes for two and three streets:

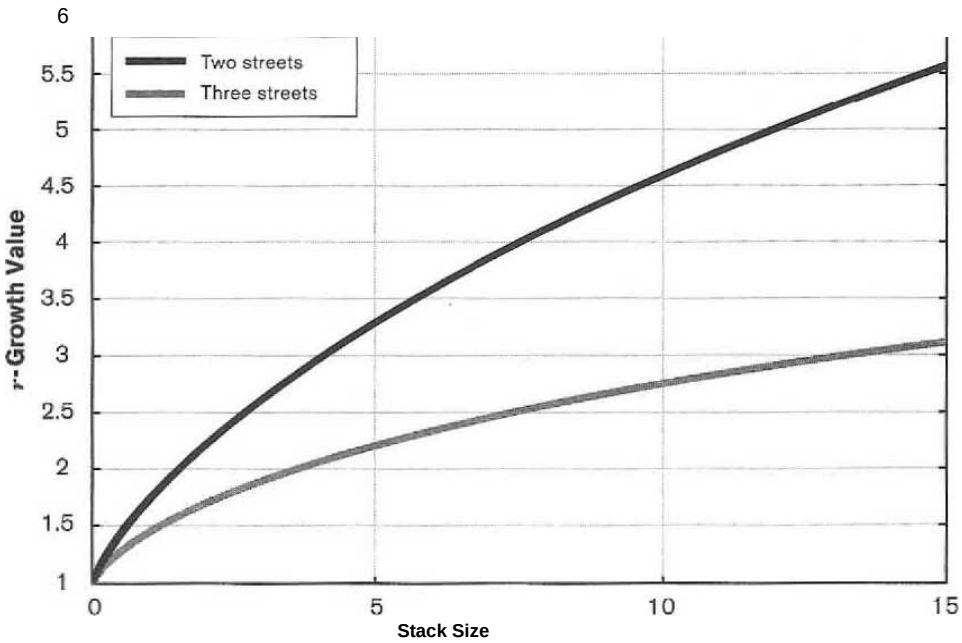

Figure 19.1. Geometric Growth of pot for two and three streets

When *r* is equal to 3, then *s* is equal to 1 and the bets are pot-sized. You can see that in order for pot-sized bets to be optimal in the three-street case, the stacks must bet 13 times the initial pot. At stacks of 3.5 times the pot, half-pot bets are sufficient to get all the money in geometrically.

Auction Games (Games with Infinite Streets)

The last static games we will consider are not precisely like poker games; in fact, the structure of their solutions is rather different from the structure of some of the preceding games. However, they do provide us with at least one important principle and have some interesting properties of their own. These are games with infinite streets (or arbitrarily large numbers of streets). We call

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these ***auction games*** as well because in the limit as the number of streets goes to infinity, they converge onto a different type of game altogether.

Example 19.4

We begin by considering a clairvoyance game of an arbitrary number of streets.

M full streets.
Static hand values.
Initial pot size of 1 unit.
No-limit betting (stacks of S units).
Y is clairvoyant.

Extending the geometric growth of pot results to M streets, we know the following will be true:

The final pot size (if all bets are called) will be 1 + 2*S* (that is, the current pot size plus both stacks).

$P = 1 + 2S$

The growth on each street will be the Mth root of P (this is the geometric growth of pot for M streets), so

$G = P^{(1/M)}$

The bet size *s* (in current pots) on each street will be:

$s = (G - 1)/2$

Suppose that we call the pot on the current street 1. Y will bet the correct number of hands for value yv and bluff with $s/(1+ s)$ times as many hands. His total betting hands, then, will be $y_v(1 + 2s)/(1 + s)$.

We can state this as follows. The ratio of the hands bet on street m+1 to the hands bet on street m will be:

$r_b = (1 +2s)/(l +s)$

You can think of this value as the "bluff multiplier." Suppose we have a two-street game with 1 unit stacks and a pot of 1. Growing the pot geometrically would mean growing the pot from 1 to 3 in two streets. So r for this game is $\sqrt{3}$ and the bet size for each street is $(\sqrt{3} - l)/2$, or about 0.366. To find the bluffs here, we simply figure that Y has *y* value bets. So on the river, he will bluff $s/(l + s)$ times *y*. The expression $s/(l + s)$ for this bet size is about 0.268. Hence, Y will bet 1.268y hands on the river, and $(1.268)^2$y hands on the turn. Substituting 0.366 for *s* in the above expression for r_b, we find that r_b is 1.268.

Suppose Y has a total number of nut hands for the entire game *y*. Let ym be the number of hands he bets on street m + 1, while y_M is the total number of value hands, or y_M = y. Also let y_0 be the number of hands he bets on the first street.

Then we have:

$y_m = y_0 rb^m$

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$y_M = y_0 rb^M = y$
$y_m = yrb^{m-M}$
$y_0 = y_{rb}^{-M}$

We may, however, run into a problem here. Suppose that $y_0 > 1$. Then we have a situation we've discussed earlier where Y doesn't have enough bluffs, since adding enough bluffs would make him bet more hands than he has. Hence, he can simply bet and claim the pot and X should optimally fold.

Otherwise, X calls with his typical frequency of 1/(1 + s). Let xm be the number of hands he calls with on the m + 1th street. Then:

$r_c = l/(l + s)$
$x_m = r_c^m$

In either of these cases, assuming that Y plays optimally, X is indifferent to calling or folding. So the value of the game is equal to y_0 - y, which is the pot for every hand Y bluffs initially.

Next we consider what seems to be an entirely different type of game:

Example 19.5 - The Auction Game

Static hand values.
Initial pot size of 1 unit.
Stacks of S units.

Each player makes a bid secretly, and the bids are simultaneously revealed.
The bids are s_x and s_y and must be on the interval [0, S]. Both bids are put into the pot.

If $s_x > s_y$, then X wins, if $s_y > s_x$, then Y wins.
If $s_y = s_x$, then there is a showdown. (This will usually occur when the bids are both S).

This game is not quite so poker-like as some other games we have examined. However, let's be a little abstract. Suppose we have a poker game with static hand values and M streets. Let's suppose we are going to play a strategy that is inspired by the auction game.

For each hand we can hold, we are going to pick a number s_x that is some fraction of our total stack 5.

On each street, we are going to bet an equal piece of our remaining stack $\varepsilon = S/M$ until we reach our threshold value s_x, after which point we will just check and give up. ε is the Greek letter epsilon, which we introduced earlier and represents a very small amount. As we shall see, in this game, epsilon will go to zero as the number of streets grows. If our opponent puts in a raise of any amount, we will call if the raise is to an amount smaller than our threshold value and begin the next street by betting ε for our remaining stack and the remaining streets. If the raise is to an amount higher than our threshold, we will fold.

Now suppose our opponent is the nemesis. How much can he gain from exploiting our strategy? In order to find out where our threshold value is. he must call the initial bets that led up to our check. Thus, he cannot effectively exploit us by calling along and bluffing us out of the pot. In fact, the only gain he can realize is when our strategy forces us to bet an amount that is slightly larger than s_x. Suppose that we have 100 streets, and s_x is 0.601. For the first sixty streets, we bet 0.01 of the initial pot. On the sixty-first street, however, we still have not reached our threshold.

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Now we cannot bet a smaller amount without exposing ourselves to exploitation. So we. again bet 0.01, making our total bet 0.61. This is 0.009 more than we were actually willing to risk, and against almost all of the hands that make it to this point, we lose this additional 0.009 of value.

We can assess the value of a raise to the nemesis as well. Suppose that we bet on a particular street, making the total amount we have bet equal to x. Since we bet this street, the nemesis knows that our sx value must be less than x - ε. Suppose he then raises to *y* units. There are three cases:

$s_x < y$. In this case, we fold. However, the nemesis could have made more money (s_x - x at least) by simply calling along until we gave up.

$s_x > y$. In this case, we call. After the raise, however, our bet size for each street has been reduced - instead of betting $(S - x)/M_{cur}$ per street the rest of the way, we will bet

$(S - y)/M_{cur}$, a smaller amount. Since we already showed how the nemesis can gain up to ε against our strategy, he certainly would not want to make it smaller.

$s_x = y$. In this case, we simply call and then check and give up on later streets. The nemesis gains nothing.

In no case can the nemesis gain anything by raising - in fact, he ought to simply call down as well.

Another way of seeing this is to consider two players who play this strategy against each other. The following is a result from game theory: if we have a strategy pair (X, Y), and we know the value of the nemesis of X and the nemesis of Y, the game value must be between those two values. This is because the optimal strategy for X must do worse than the nemesis against Y, and vice versa. Hence, the value to the nemesis of our employing this betting structure instead of the optimal structure is at most the value of the game played optimally G - ε.

We know that as the number of streets M approaches infinity, ε will go to zero, because M is the denominator of the expression for ε = S/M. So we know that the value of any game converges on the value of the auction game that corresponds to the parameters of that game. Likewise, we know that for any number of streets, playing an auction strategy gives up at most ε, or S/M.

Example 19.6

Next, we consider the clairvoyant auction game.

Y is clairvoyant.
Auction game rules with stacks of *S* and initial pot 1.

Let Y have *y* winning hands. Then there are three types of hands total: X's hands, which are all alike and Y's hands, which are winning hands and losing hands. Y will evidently always choose $s_y = S$ when he has a winning hand; that is, Y will always be willing to bet his entire stack when he is going to win the pot. Let *y* be the amount of winning hands that Y has. Let $x(s)$ be the fraction of hands with which X bids at least *s*, and $y(s)$ be the fraction of hands with which Y bids at least *s*.

Y's strategy will be to bid *S* with nut hands, and to choose s_y for bluffing hands such that he gradually gives up on bluffing hands until he is left with $y(S) = y$. To make Y indifferent between bidding two amounts *s* and $(s + \Delta s)$ with a dead hand, we have the following. Δs, or

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"delta s" is generally used to denote changes (small or large) in s. So here Y is indifferent between bidding s and bidding an amount greater by Δs.

$-\Delta x = x(s + \Delta s) - x(s)$
(We use this notation became as s increases, x will decrease).

$-x\Delta s + \Delta x\Delta s + (1 + 2s)(-\Delta x) = 0$
$(1 + 2s + \Delta s) - \Delta x = x\Delta s$
$-\Delta x / x = \Delta s / (1 + 2s - \Delta s)$

as $\Delta s \rightarrow 0$, we have:
$-\Delta x / x = \Delta s / (1 + 2s - \Delta s)$
$-\Delta x / x \approx \Delta s / (1 + 2s)$
$-dx/x = ds/(1 + 2s)$

Integrating both sides, we have:

$-\ln(x) + C = \ln(l + 2s)/2$
$\ln(Cx^2) = \ln(l + 2s)$
$x = k / \sqrt{(1 + 2s)}$

Of course, x(0) = 1 as all hands must bid at least zero. Hence k = 1, and we have:

$x(s) = 1/ / \sqrt{(1 + 2s)}$

In the tame way, Y must male X indifferent between s and $s + \Delta s$:

$(-\Delta y)(1 + 2s + 2\Delta s) + y(\Delta s) = 0$
$-\Delta y/y = \Delta s/ (1 + 2s + 2\Delta s)$

As $\Delta s \rightarrow 0$:

$dy/y = ds/(l + 2s)$
$y(s) = k/ \sqrt{(1 + 2s)}$

We know that $y(S) = y_0$, so:

$y = k/ \sqrt{(1 + 2s)}$

So $y(s) = y\sqrt{\frac{1+2S}{1+2s}}$ and $x(s) = 1/\sqrt{(1 + 2s)}$

However in this case, too, y_0 may be greater than 1, as we saw when looking at the *x(s)* values

So the solution to the game is:

Solution:

$y(s) = y\sqrt{\frac{1+2S}{1+2s}}$ (or 1, whichever is smaller)
$x(s) = 1/\sqrt{(1 + 2s)}$ (or 0 if $y(\sqrt{(1 + 2s)}) > 1$)

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To recap what these solutions mean, these are the total fraction of hands with which X and Y will bid at least *s* in the auction (if their entire stack is *S)*. Suppose that the stacks are 10 total and Y has a winner 50% of the time. Then Y will bid at least 1 unit with:

$$y(1) = (0.5)(\sqrt{(1+2(10))/(1+2(1))})$$
$$y(1) = (0.5)(\sqrt{7})$$
$$y(1) \approx 1.323$$

Since this value is greater than one, Y will bid at least 1 with all hands. He wilt bid at least 5 with:

$$y(5) = (0.5)(\sqrt{(1+2(10))/(1+2(5))})$$
$$y(5) = (0.5)(\sqrt{(21/11)})$$
$$y(5) \approx 0.691$$

Here Y will bid at least 5 with 0.5 winning hands and 0.191 of his hands as "bluffs," and so on.

We can also show that these solutions are the limit as $M \rightarrow \infty$ of the clairvoyant poker game with *M* streets. We will ignore here the cases where Y lacks enough bluffs, and concentrate on the more usual case where both sides have mixed strategies on the first street.

To avoid confusion with regard to the values of *s*, let

$s_n = (G - 1)/2$

which is the one-street betting ratio. (See Example 19.4)

We can use the following approximations:

$s_n = (G - 1)/2 \approx \ln(G)/2 = \ln P/ 2M$

We know from the solution to the clairvoyant poker game (Example 19.1) that $x_m = r_c^m$.

$$x_m = r_c^m$$
$$x_m = (1+s_1)^{-m}$$
$$x_m \approx e^{-ms1}$$
$$x_m \approx \exp(-\frac{m}{2M} lnP)$$
$$x_m \approx P^{-\frac{m}{2M}}$$

We also know that the pot size after this street will be:

$$1 + 2s_1 = G_m$$
$$1 + 2s_1 = P^{\frac{m}{M}}$$

Substituting, we have:

$$x_m = (1+2s_1)^{-1/2}$$
$$x_m = x(s)$$

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We can show that the same holds true for y(s) - that the solution for the auction game in the limit as M goes to infinity is the same as the clairvoyant no-limit game with *M* streets.

Key Concepts

- The clairvoyant game can be extended to more than one street. The key feature of the multi-street game is that the clairvoyant player carries forward a set of hands from street to street, bluffing many more hands on earlier streets and giving up on a fraction of those hands on each street.
- In the no-limit multi-street clairvoyant game, the optimal bet size is the geometric growth of pot - growing the pot by the same fraction on each street.
- As the number of streets increases, positional value goes to zero - the auction game is symmetric.
- As the auction game (and geometric growth of pot) show, it is optimal to bet small fractions of the pot in static games. Non-static games do not necessarily follow this principle.
- Auction strategies assign a value s_x, to each hand. This is a reasonable way to think about no-limit games in general, as having a distribution of hands that each has a value. Then the goal is to correlate the pot size with our hand strength.
- The non-clairvoyant player plays each street independently as if it were a single street game,
- The hands that Y bluffs once and then gives up on have the *a* bluffing ratio relationship to the total hands that Y will bet for value and bluff on the second street.

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